



Mechanism of azo dye degradation in Advanced Oxidation Processes: Degradation of Sulfanilic Acid Azochromotrop and its parent compounds in aqueous solution by ionizing radiation

Tamás Pálfi, László Wojnárovits, Erzsébet Takács*

Institute of Isotopes, Hungarian Academy of Sciences, H-1525 Budapest, P.B. 77, Hungary

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ABSTRACT

Mechanistic studies were made on hydroxyl radical and hydrated electron reaction with Sulfanilic Acid Azochromotrop (SPADNS) as model azo dye in dilute aqueous solution. SPADNS contains 4,5-dihydroxynaphthalene-2,7-disulfonic acid part and 4-sulfophenylazo group. To establish the details of the reaction mechanism the reactions of two simpler molecules without 4-sulfophenylazo part were also studied: one of them contained one (in position 4, II), the other two (in positions 4 and 5, III) –OH groups. Hydroxyl radicals react with these molecules with radical addition to the naphthalene-2,7-disulfonic acid part. The adduct hydroxycyclohexadienyl type radical decays in radical–radical reactions, or undergoes a (pH dependent) water elimination to yield naphthoxy radical. The radical decay takes place on the ms timescale. Degradation efficiencies are 0.6–0.8. Hydrated electron in the case of the two simpler molecules reacts with the rings, while in the case of dye with the azo bond. Electron scavenging is followed by protonation, this reaction in the case of II and III yields cyclohexadienyl, while with the dye hydrazo radical. The efficiency of degradation with II and III is 0.2–0.6, while for SPADNS it is close to 1.

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1. Introduction

Azo dyes are considered to be hazardous materials because of their slow biological degradation and of their harmful intermediate degradation products. For these reasons their decomposition in Advanced Oxidation Processes is under intensive research. Electron beam (EB) treatment belongs to the class of Advanced Oxidation Processes (AOP): here also, as in most of AOP principally $\cdot\text{OH}$ radicals induce the chemical transformations. Laboratory investigations, pilot-plant experiments and an industrially established technology illustrated that EB treatment can efficiently destroy textile dyes in aqueous solutions (Han et al., 2002; Paul et al., 2010a, b).

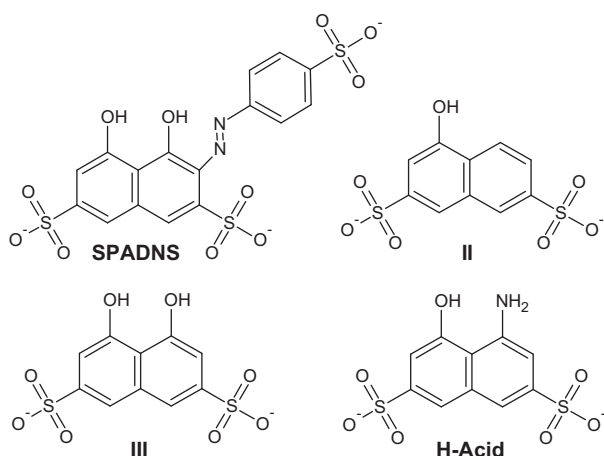
Most of the studies in the field are concerned with the degradation of the starting molecule following the degradation by spectrophotometry, or by HPLC separation and detection. The mechanism of degradation was much less frequently studied (Arslan-Alaton et al., 2009).

Here we report on mechanistic studies choosing Sulfanilic Acid Azochromotrop (4,5-dihydroxy-3-(4-sulfophenylazo)naphthalene-2,7-disulfonic acid, SPADNS) as model dye (Scheme 1). This molecule contains a 4,5-dihydroxynaphthalene-2,7-disulfonic acid part and, connected to it, in position 3 a 4-sulfophenylazo group. In order to

establish the details of the reaction mechanism we studied the reactions of the naphthalene unit on the example of two simpler molecules: 4-hydroxynaphthalene-2,7-disulfonic acid (II) and 4,5-dihydroxynaphthalene-2,7-disulfonic acid (III). In the evaluation of the structural relationships we use also the results published on H-acid (4-hydroxy-5-aminonaphthalene-2,7-disulfonic acid) in our former work (Pálfi et al., 2007). In the structure of H-acid an –OH group and an –NH₂ group is on the naphthalene-2,7-disulfonic acid in 4 and 5 positions. II, III and H-acid are central parts of many azo dyes.

In the present work the reactive radical intermediates that induce the degradation of solute were generated by the radiolysis of water. In irradiation wastewater treatment it is generally assumed that only the hydroxyl radical ($\text{HO}\cdot$, yield $0.28\ \mu\text{mol J}^{-1}$) intermediate of water radiolysis takes part in the degradation of the organic pollutant. The other two reactive intermediates, hydrated electron and hydrogen atom (e_{aq}^- and $\text{H}\cdot$, yields 0.28 and $0.055\ \mu\text{mol J}^{-1}$, respectively) in reaction with dissolved oxygen molecules quickly (rate coefficient $\sim 2 \times 10^{10}\ \text{mol}^{-1}\ \text{dm}^3\ \text{s}^{-1}$) transform to the $\text{O}_2^{\cdot-}/\text{HO}_2^{\cdot}$ pair (Swallow, 1982). The latter species are rather inactive in their reaction with dye molecules. However, the dye molecules similarly to O_2 also react with hydrated electrons with rate coefficients of $\sim 10^{10}\ \text{mol}^{-1}\ \text{dm}^3\ \text{s}^{-1}$, hence when the dye concentration is comparable with the dissolved oxygen concentration ($\sim 2.5 \times 10^{-4}\ \text{mol dm}^{-3}$ in air saturated solution), e.g. in the wastewater of a textile factory, the e_{aq}^- reactions may not be

* Corresponding author. Tel.: +361 392 2612; fax: +361 392 2548.
E-mail address: takacs@iki.kfki.hu (E. Takács).



Scheme 1. Structural formulas of the compounds studied.

negligible. We study also the e_{aq}^- reactions and compare them with the $HO^\bullet$ reactions. In order to observe the first steps of degradation, all experiments were made under oxygen-free conditions: oxygen by reacting with the dye radicals complicates the observation of primary processes (Wojnárovits and Takács, 2008; Földváry and Wojnárovits, 2009).

2. Experimental

2.1. Chemicals

SPADNS, compounds II and III were obtained from Sigma-Aldrich. Their structures and stabilities were checked by NMR and FTIR spectroscopy. Usually 0.5 mmol dm^{-3} concentration solutions were used in the experiments. Perchloric acid and sodium hydroxide were applied to set the pH of the solutions. Methanol (HPLC grade) and tetrabutyl-ammonium-hydrogen sulphate (TBAHS) used as HPLC eluents were obtained also from Sigma-Aldrich, *tert*-butanol from Spektrum-3D.

2.2. Protonation

The protolytic equilibrium constants were determined by potentiometric measurements using pHCal software (Peintler et al., 2007) at constant ionic strength (0.1 mol dm^{-3} NaCl). This program needs a careful calibration of the pH meter in case of complex equilibrium.

2.3. Irradiation

Deoxygenation of solutions was carried out by bubbling with appropriate gases for 25 min before irradiation and the bubbling was continued during the experiments. When the e_{aq}^- reactions were studied the N_2 purged solutions contained 0.5 mol dm^{-3} *tert*-butanol in order to transform the reactive $HO^\bullet$ radicals to unreactive *tert*-butanol radicals (reductive conditions). The reactions of $HO^\bullet$ radicals were studied in N_2O bubbled solutions; in such solution e_{aq}^- is converted to hydroxyl radical (oxidative conditions).

The γ -radiolytic experiments were carried out using a ^{60}Co irradiation facility with dose rate 3.0 kGy h^{-1} as determined by Fricke dosimeter. Solutions were analysed before and after reaction by using Jasco 550 UV–vis spectrophotometer and by HPLC system for separation.

The transient intermediates were studied by pulse radiolysis. The set-up is working with 800 ns pulses of 4 MeV electrons. The

time dependent changes in the absorbance were followed with UV–vis kinetic spectrophotometry in 2 cm cells (Földiák et al., 1988). The dose/pulse values were generally between 5 and $30 \text{ Gy (J kg}^{-1}\text{)}$ determined by thiocyanate dosimeter.

In calculating the absorption spectra of transients, due to the high depletion, a correction was applied for the decomposed dye molecules using the equation developed previously (Wojnárovits et al., 2005; Wojnárovits and Takács, 2008):

$$\sum_{i=1}^n \varepsilon_i^t c_i^t = \varepsilon_S^t (c_S^{t=0} - c_S^t) + \frac{\Delta A^{t,\lambda}}{l} \quad (1)$$

In the equation $c_S^{t=0}$ and c_S^t are the dye concentrations at $t=0$ and t , c_i^t is the intermediate concentration at t , ε_i^t and ε_S^t are the molar absorption coefficients of the dye and the intermediate, $\Delta A^{t,\lambda}$ is the measured Δ absorbance at time t , l is the optical cell length.

2.4. HPLC system

The chromatographic system consisted of a Jasco PU-2089Plus quaternary gradient pump, a Jasco MD-2015Plus diode-array Multiwavelength Detector and a Nucleosil 100 C18 column (Teknokroma[®]; pore size $5 \mu\text{m}$, length 25 cm, diameter 0.4 cm). Ion-pair separations were made using 50 mmol dm^{-3} aqueous solution of tetrabutyl-ammonium-hydrogen sulphate (TBAHS) at pH 6.1 and methanol in 60:40 ratio. All mobile phases were filtered using a Millipore $0.45 \mu\text{m}$ filter and were degassed by ultrasonication before use. The separations were made at room temperature using injection volume of $10 \mu\text{l}$, and flow-rate of $0.8 \text{ cm}^3 \text{ min}^{-1}$.

2.5. Quantum chemical calculations

A Gaussian 98 program package was used in the calculations with a HF/6-311G** base (Frisch et al., 2002), while for analyzing the optimized structures and for making drawings ChemCraftTM software was applied (Zhurko, 2008).

2.6. Kinetic analysis

Fittings were carried out by the quaternary Runge–Kutta method (Press et al., 1992) using Microsoft Excel. The rate coefficients were calculated by solving the appropriate differential equations.

3. Results and discussion

3.1. pK_a 's of the compounds

The compounds investigated are strong acids: $-\text{SO}_3\text{H}$ group deprotonates below pH 1. The protonation of molecule II at the $-\text{OH}$ group is characterised by $pK_a=7.96$. Similar pK_a value (8.41) was determined previously for the H-acid (Table 1). The values for II and III molecules, however, differ considerably.

Table 1
 pK_a -values of the compounds investigated.

Compound	pK_{a0}	pK_{a1}	pK_{a2}
4,5-dihydroxy-3-(4-sulfophenylazo)naphthalene-2,7-disulfonic acid (SPADNS)		8.75	12.16
4-hydroxynaphthalene-2,7-disulfonic acid (II)		7.96	
4,5-dihydroxynaphthalene-2,7-disulfonic acid (III)		5.06	11.66
4-amino-5-hydroxy-2,7-naphthalenedisulfonic acid (H-acid) ^a	3.29 ^b	8.41	

^a Pálfi et al., 2007.

^b Deprotonation/protonation at the amino group.

The first protonation of molecule III is observed at lower, that of the second one at higher pH than that of molecule II (Table 1, Scheme 2). There are hydrogen bonds in both, the double protonated and single protonated forms (Inskip et al., 1968): according to our quantum chemical calculations in the case of the doubly protonated molecule the structure (both parallel and antiparallel) with hydrogen bond is more stable by $\sim 27 \text{ kJ mol}^{-1}$ energy than the structure without hydrogen bond. Due to the hydrogen bond all atoms in the six-membered ring containing the hydrogen bridge, except oxygens on $-\text{SO}_3^-$, are in the same plane. The shifts in the pK_a values are due to the hydrogen bond between the two $-\text{OH}$ groups: the hydrogen bond promotes the dissociation of the first proton and hinders the dissociation of the second proton.

In the case of SPADNS the azo/hydrazone tautomerism also complicates the protolytic equilibrium. In aqueous solutions of azo dyes usually the hydrazone form dominates (Bredereck and Schumacher, 1993): our spectrophotometric investigations with SPADNS also suggest the dominance of the hydrazone form. The absorbance maximum of the hydrazone form is at lower wavelength as compared with that of the azo form (Abbott et al., 2005).

With potentiometric measurements we determined pK_a 8.75 and 12.16 for SPADNS: similarly to molecule III, here also there is large difference between the two pK_a values. Based on the literature analogies the first pK_a value may belong to protonation at the $-\text{OH}$ group not participating in the azo/hydrazone tautomerism. With other azo dyes the $-\text{OH}$ group participating in the tautomerism was found to dissociate at $\text{pH} > 12$.

There is an enormous change in absorption properties of SPADNS molecules as the pH passes through pK_a 8.75 (Fig. 1). As a result of the first deprotonation the absorbance maximum in the visible range shifts to longer wavelength and the absorbance value at the maximum highly decreases. Based on the enormous change we assume that after deprotonation the ion exists in the azo form as it is shown in Scheme 3. The release of the second proton with pK_a value 12.16 does not imply spectral changes since the deprotonation of the second $-\text{OH}$ group just slightly influences the electron conjugation through the azo bond.

3.2. Reactions of $\text{HO}^\bullet$ radical

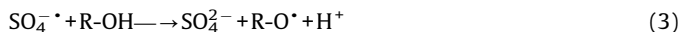
In the reactions of hydrated electron, showing strong reductive property, and that of the hydroxyl radical with oxidative property, different organic molecule intermediates are expected. In the case of hydroxyl radical reaction the energy of the highest occupied molecule orbital (HOMO) and its distribution determines the rate coefficient and the place of radical attack (Özen et al., 2003).

With molecules II and III, and also with SPADNS the HOMO orbitals are concentrated on the rings; therefore the $\text{HO}^\bullet$ radical reactions are basically influenced by the electron distribution on the ring (Fig. 2). The $\text{HO}^\bullet$ radicals react with all the molecules investigated with diffusion limited rate coefficient, $\sim 1 \times 10^{10} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$.

In the transient spectrum of II obtained by pulse radiolysis there are two main absorption bands at 305 and 325 nm (Fig. 3). The spectrum of molecule III, which has two $-\text{OH}$ groups, contains three further intensive bands at 330, 345 and 360 nm and at pH 3.56 there is a long tail towards the infrared region (Fig. 4). The

360 nm band was also observed in the absorption spectrum of H-acid (Pálfi et al., 2007).

In the $\text{HO}^\bullet$ radical reactions of phenol derivatives generally two types of radicals, hydroxycyclohexadienyl type radicals (with radical site on the ring) and phenoxy (naphthoxy) radicals (radical site on the O atom of a hydroxyl group) are the intermediates (Von Sonntag, 2006). In order to observe naphthoxy radicals we studied the transient products of reactions of II with sulphate radical anion ($\text{SO}_4^{\bullet-}$). In the experiments 0.05 mol dm^{-3} persulphate was given to the solution, in such solution e_{aq}^- is converted to $\text{SO}_4^{\bullet-}$ in reaction (2) (Hayon et al., 1972). The reaction of phenolic compounds with this radical yields phenoxy radicals in electron transfer reaction (3) (Neta et al., 1977)



In the transient spectrum of naphthoxy radicals two main bands were identified with maxima at 355 nm ($\epsilon_{355} = 10000 \text{ mol}^{-1} \text{ dm}^3 \text{ cm}^{-1}$) and at 400 nm ($\epsilon_{400} = 4800 \text{ mol}^{-1} \text{ dm}^3 \text{ cm}^{-1}$) (Fig. 3, Table 2), however, at longer wavelengths there is a wide structureless absorption band which continues up to the highest wavelength point measured here.

The phenoxy (naphthoxy) radicals characteristically have three absorption bands (Johnston et al., 1993; Steenken and Neta, 2003): an intense band between 300–400 nm (depending on the parent molecule) a weaker and often structured band in the 350–420 nm range ($\epsilon_{\text{max}} \approx 4000 \text{ mol}^{-1} \text{ dm}^3 \text{ cm}^{-1}$; Roder et al., 1999) and a weak, ill-defined wide band in the 500–700 nm range. Such radicals generally decay in slow second order reactions (as observed here). The spectrum obtained in $\text{SO}_4^{\bullet-}$ reaction of molecule II shows these characteristics: in the 355–580 nm range the spectrum has

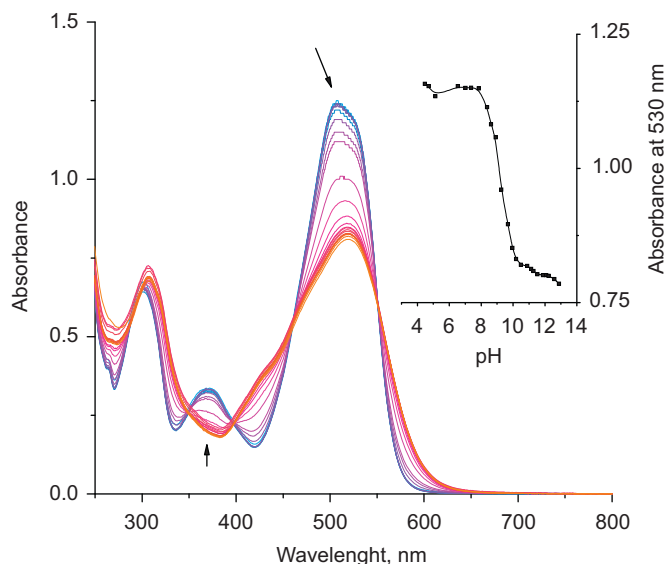
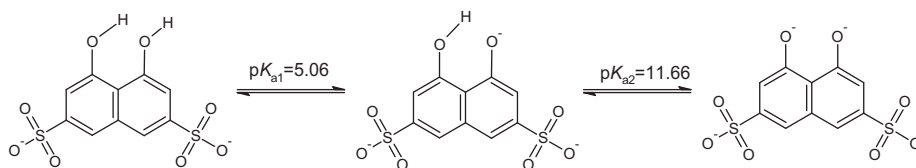
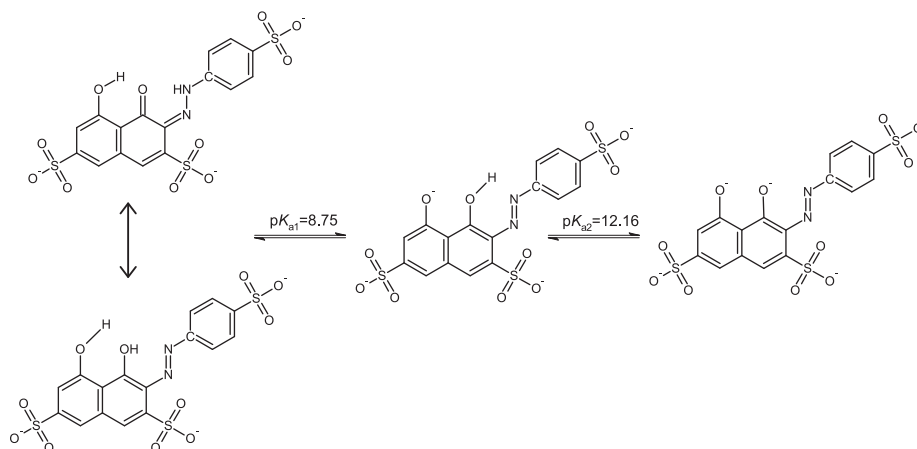


Fig. 1. Absorption spectra of $4 \times 10^{-5} \text{ mol dm}^{-3}$ solution of SPADNS in a wide pH range. The arrows show the pH dependent changes. Inset: pH dependence of absorbance at 530 nm.



Scheme 2. Protolytic equilibrium of molecule III.



Scheme 3. Azo-hydrazo tautomerism and protolytic equilibria of SPADNS.

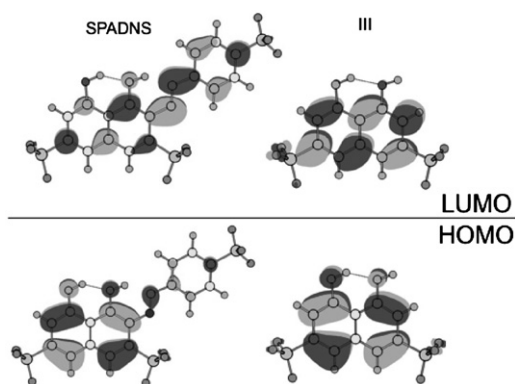


Fig. 2. LUMO and HOMO orbitals of SPADNS and molecule III as obtained by HF/6-311G* calculations.

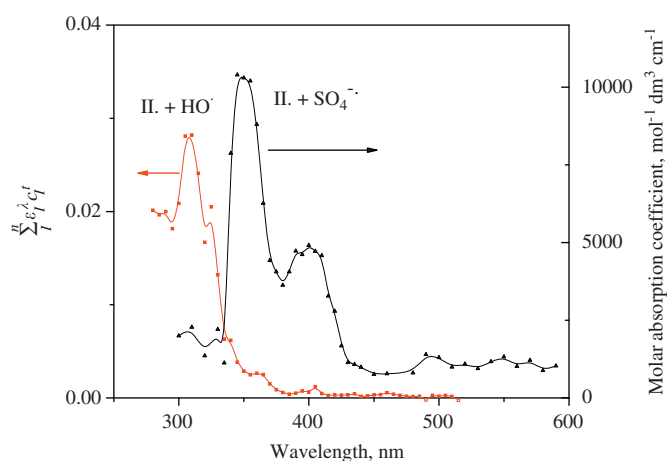


Fig. 3. Absorption spectra of transients formed in $\text{HO}^\bullet$ radical reactions with II ($t=10\ \mu\text{s}$, $c=0.1\ \text{mmol dm}^{-3}$, $\text{pH } 4.5$, $D=10\ \text{Gy imp}^{-1}$) and in the $\text{SO}_4^{\bullet-} + \text{II}$ reaction.

considerable absorbancies, while the absorbance of intermediates formed in $\text{HO}^\bullet$ reaction is weak in this range. We can conclude that naphthoxy radical formation in $\text{HO}^\bullet$ reaction with molecule II is not probable.

In contrast to molecule II, in case of molecule III the spectral characteristics at pH 3.56 in the 400–600 nm range, reflect the presence of naphthoxy radicals (Fig. 4). These radicals are suggested to form in $\text{HO}^\bullet$ radical addition to the ring and subsequent water

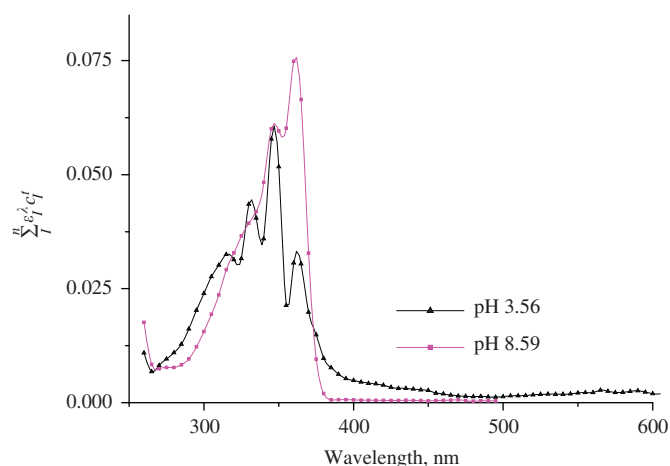


Fig. 4. Spectra of intermediates formed in $\text{HO}^\bullet + \text{III}$ reaction at pH 3.65 and 8.59, $t=10\ \mu\text{s}$, $c=0.1\ \text{mmol dm}^{-3}$, $D=10\ \text{Gy imp}^{-1}$.

elimination (as it is shown for SPADNS in Scheme 4). Naphthoxy radical formation is expected to be pH dependent. By increasing the pH in solution of molecule III one of the $-\text{OH}$ groups deprotonates with pK_a 5.06. In the transient spectrum the 345 nm absorbance remains unchanged, while the absorbance of the band with maximum at 365 nm increases by a factor of two (Fig. 4, Table 3). The radical lifetime remains unaltered. It is worth mentioning that in the absorption spectrum of the starting molecule there is a similar large change caused by deprotonation. Deprotonation of one of the $-\text{OH}$ groups may hinder water elimination and by that naphthoxy radical formation, because the single $-\text{OH}$ hydrogen atom, remained after deprotonation, belongs to both $-\text{OH}$ groups. Based on this effect we identify the 365 nm peak in the transient spectrum of III as belonging to the cyclohexadienyl type radical. The absence of considerable absorbance above 385 nm at pH 8.59 in the transient absorption spectrum of the intermediate produced in $\text{III} + \text{HO}^\bullet$ reaction also supports this identification.

Due to the structural similarities we may expect similar mechanism also for the $\text{HO}^\bullet + \text{SPADNS}$ reaction as for the $\text{HO}^\bullet$ reaction with the simpler molecules. However, the literature analogies suggest also $\text{HO}^\bullet$ radical attack on $-\text{N}=\text{N}-$ group (Wojnárovits and Takács, 2008).

In the transient spectrum of SPADNS (not shown here) we found an intensive absorption band at 365 nm similarly to compound III: this band should belong to hydroxycyclohexadienyl type radical.

The second order decay of the radical formed from SPADNS is two orders of magnitude slower ($6.0 \times 10^6 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$) than with the simpler molecules. The increased lifetime due to the higher stability of the radical is quite reasonable based on the larger molecule size.

Using the accumulated results we interpret the $\text{HO}^\bullet$ radical induced reactions as follows: the absorption band in the 300–330 nm range with molecules II, III and SPADNS belongs to the hydroxycyclohexadienyl type radical formed with $\text{HO}^\bullet$ radical addition to the naphthalene unit. In case of II this is the most intensive band, the intensity maximum is at 305 nm. After the radical addition some of the radicals produced may transform to naphthoxy radical by water elimination, this reaction with molecule II is not probable. However, in the case of molecule III the absorption spectrum (Fig. 4) indicates naphthoxy radical generation. The water elimination probably takes place in case of hydroxycyclohexadienyl radicals with two neighbouring hydroxyl groups as shown in Scheme 4.

Irradiating solutions of the compounds studied here under oxidative conditions with γ -rays the UV–vis absorption spectrum of the solution shifts towards longer wavelengths (Fig. 5). There is a gradual disappearance of the starting compound as we found it by HPLC separation and optical detection (Fig. 6).

As mentioned in the Introduction we know the yield of hydroxyl radicals (in N_2O saturated solution $0.56 \mu\text{mol J}^{-1}$) produced in the solution during irradiation. Using the analytical results of HPLC separation and detection on the amount of molecules decomposed, the efficiency and the yield of reactions that do not lead to decomposition (e.g. re-formation) can be calculated. The efficiency shows the number of molecules decomposed when one hydroxyl radical was scavenged.

At very low dose, when the accumulation of the products in solution is negligible, the time (dose) dependence of concentration

($[A_1]_0$ at $t=0$, $[A_1]$ at t) of the initial molecules can be described by a linear equation (Wojnárovits et al., 2007; Wojnárovits and Takács, 2008)

$$[A_1] = [A_1]_0 - \frac{r}{D_T} D_t \quad (4)$$

$$r = f G_T D_T \rho \quad (5)$$

D_t (Gy) is the dose accumulated till time t (s), f is the efficiency, G_T is the radiation chemical yield (J kg^{-1}), D_T (Gy s^{-1}) is the dose rate and ρ (kg dm^{-3}) is the density, r ($\text{mol dm}^{-3} \text{ s}^{-1}$) and $r' = r/D_t$ ($\text{mol dm}^{-3} \text{ Gy}^{-1}$) are the rates of degradation. Using the calculated and measured rates we can estimate the efficiency of degradation. The efficiency of degradation (f) in $\text{HO}^\bullet$ reactions for the two simpler molecules is smaller than for the two dye molecules (Table 3).

As mentioned before, one of the main intermediates is the hydroxycyclohexadienyl radical; two of these radicals may react in disproportionation reaction. From one of the radicals such product forms in which one of the double bonds in the rings is saturated, the other radical transforms to a molecule in which the original aromatic structure is re-established. However, the latter molecule contains an extra $-\text{OH}$ group; this extra substituent causes a red

Table 3
Degradation rates (r' and r) and efficiencies (f) in N_2O saturated solutions at pH 5.5.

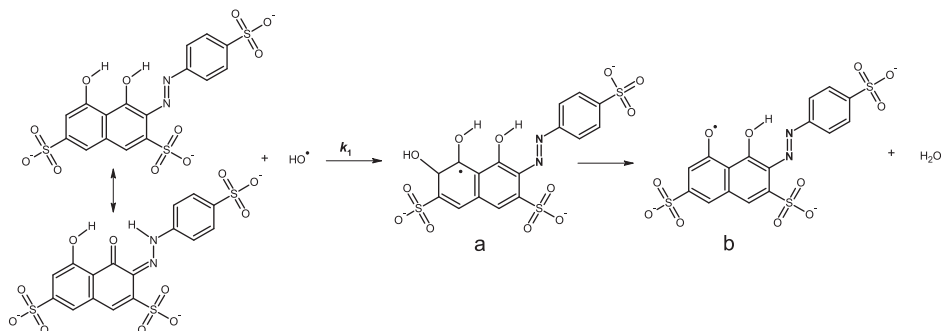
	r' ($\text{mol dm}^{-3} \text{ Gy}^{-1}$)	r ($\text{mol dm}^{-3} \text{ s}^{-1}$)	f
II	3.86×10^{-7}	3.22×10^{-7}	0.69
III	3.13×10^{-7}	2.61×10^{-7}	0.56
SPADNS	4.28×10^{-7}	3.56×10^{-7}	0.77
AR-28 ^a	4.07×10^{-7}	3.39×10^{-7}	0.73

^a Apollonix Red azo dye taken from Pálfi et al. (2007).

Table 2
Kinetics and optical characteristics of transients formed in reactions between $\text{HO}^\bullet$ and molecules II and III, respectively. k_1 is the formation and k_2 is the decay rate coefficient.

Compound	II		III	
	$\text{HO}^\bullet$	$\text{SO}_4^{\bullet -}$	$\text{HO}^\bullet$	
pH	5.80	5.8	3.56	8.59
λ_{max} (nm)	305 (4600)		315 (5800)	
ϵ_{max} ($\text{mol}^{-1} \text{ dm}^3 \text{ cm}^{-1}$) ^a	325 (3100)		330 (7800)	~330 (7000)
			345 (10500)	345 (10700)
	360 (260)	355 (10000)	365 (5500)	365 (12000)
	405 (220)	~400 (4000)		
		~500 (1350)	565 (500)	
k_1 ($\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$)	$\sim 1 \times 10^{10}$	350: 9.5×10^9 405: $\sim 1 \times 10^{10}$	$\sim 1 \times 10^{10}$	$\sim 1 \times 10^{10}$
λ (nm), k_2 ($\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$)	305: $> 1.5 \times 10^9$ 325: $> 1.6 \times 10^9$	350: 2.8×10^8 405: 5.0×10^8	365: 4×10^8 450: 5×10^8	

^a Molar absorption coefficients (ϵ_{max}) were calculated assuming single intermediate for each reaction.



Scheme 4. Hydroxycyclohexadienyl (a) and naphthoxy (b) radical formation in SPADNS molecule.

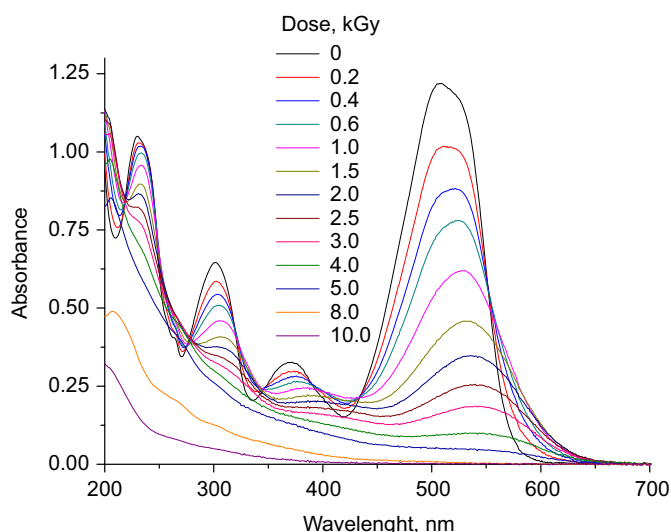


Fig. 5. Effect of $\text{HO}^\bullet$ reaction on the absorption spectra measured in N_2O saturated SPADNS solution, $c=5 \text{ mmol dm}^{-3}$, optical length 0.1 cm.

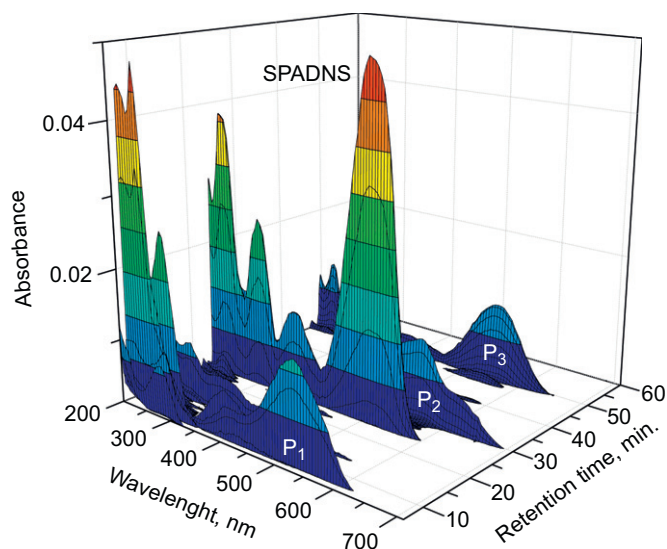


Fig. 6. Chromatogram of N_2O saturated of $5 \times 10^{-3} \text{ mol dm}^{-3}$ SPADNS solution irradiated with absorbed dose 1.2 kGy, pH 6.7, eluent 40/60 methanol/TBAH solution, flow rate 0.8 ml min^{-1} . P_1 , P_2 , and P_3 are the degradation products.

shift in the absorption spectrum. In the reaction of two naphthoxy radicals dimer molecules with conjugated electron systems are expected. The absorption spectra of these molecules are also expected to be red-shifted as compared with the spectra of the starting dye molecule.

In the case of dye molecules, however, there is a faster transformation, and the efficiency of degradation is higher than in the case of simpler molecules. Based on the increased rate we assume $\text{HO}^\bullet$ radical addition also to the azo bond and direct azo bond degradation. However, the basic primary process of $\text{HO}^\bullet$ radical action is the reaction with the aromatic ring system and not with the azo bond.

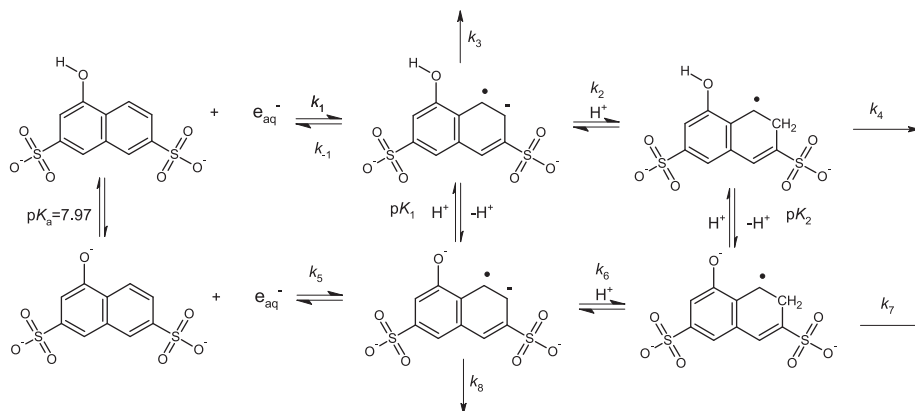
3.3. Reactions of e_{aq}^-

The rate coefficient of hydrated electron reaction is high with molecules possessing low lying un-occupied molecule orbital (LUMO): according to the molecule orbital theory the scavenged electron is accommodated on this orbital (Swallow, 1982). In molecules II and III the LUMO orbital is concentrated on atoms of the aromatic rings independently of the protonation form of the molecule (Fig. 2): for these molecules we expect hydrated electron addition to the ring. The scavenged electron is delocalized on the aromatic conjugated system; therefore, in the first step we may expect a single product (no regioselectivity). Although the quantum chemical results calculated for the gas phase may give energy ratios that differ from the liquid phase ratios, the calculated LUMO energies of II, 10.5782 eV and III 10.4514 eV, suggest slower product formation rate for II than for III. A possible faster re-formation of II (k_{-1}) may give the same result (Scheme 5).

In hydrated electron reaction the upper value of the rate coefficient (diffusion limited rate coefficient) is $\sim 3 \times 10^{10} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ (Wojnárovits and Takács, 2008). Based on the product build-up, e_{aq}^- reacts with compound III and SPADNS with rate coefficients close to the diffusion controlled limit. The rate coefficient for compound II is somewhat smaller, as it was predicted based on the calculated LUMO energies. Molecules of II and III differ only by an $-\text{OH}$ group. In the reaction of e_{aq}^- with II and III, intermediates with similar absorption spectra form (Fig. 7) suggesting the production of similar intermediates.

The electron addition to the ring is followed by protonation. This addition and protonation for both II and III brings about semi saturation of one of the double bonds by an H atom in the naphthalene unit yielding cyclohexadienyl type radical (Scheme 5, in $\text{HO}^\bullet$ addition hydroxycyclohexadienyl type radical forms).

The intermediates may decay in several competitive reactions. According to the experimentally obtained decay curves the



Scheme 5. Hydrated electron reaction with molecule II.

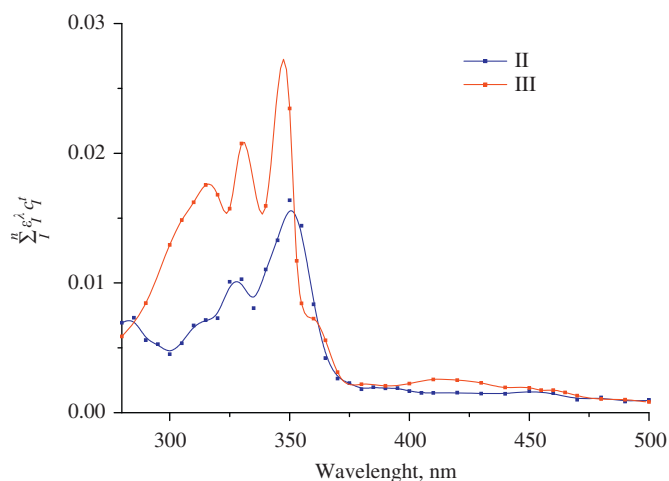


Fig. 7. Absorption spectra of intermediates produced in $e_{aq}^- + \text{II}$ and $e_{aq}^- + \text{III}$ reactions, pH 5.5, $t=20 \mu\text{s}$, $c=0.5 \text{ mmol dm}^{-3}$, $D=10 \text{ Gy imp}^{-1}$.

transient intermediates mostly disappear in 10^{-3} s . The decay of the intermediates occurs in radical–radical disproportionation and combination reactions based on literature analogies (Wojnárovits and Takács, 2008). In combination reaction a dimer molecule forms, in both subparts of the dimer one of the double bonds in naphthalene unit is saturated.

Until now we assumed that during pulse radiolysis in e_{aq}^- reaction a single intermediate forms, however, it is not precisely true. In the solution usually there are both protonated and unprotonated starting molecules and also protonated and unprotonated intermediates determined by the equilibrium constant. Hence the number of intermediates increases to 4 as it is shown in Scheme 5: the absorption spectrum and the lifetime of intermediates are expected to depend on the pH of the solution. The reactions characterised by rate coefficients k_3 , k_4 , k_7 and k_8 yield stable final products, however, these reactions may also reform the starting molecule.

The build-up and decay curves do not reflect high pH effect in the transformation of the intermediates to final products. The lifetimes of intermediates does not either differ considerably. It seems that the effect of deprotonation is larger in the case of the first deprotonation than in the case of the second, the lifetime of the electron adduct of the first deprotonated form was found three times higher than that of the protonated form.

The azo dyes have strong absorbancies in the visible spectral range, during the reaction with e_{aq}^- there is strong decoloration due to destruction of conjugation through the azo bond. Similarly to calculations with other dyes (Zhang et al., 2005) our results for SPADNS on Fig. 2 also show that the LUMO orbital is basically on the two N atoms strongly suggesting hydrated electron attack on this part of the molecule. The electron adduct formed undergoes a fast protonation as it was generally observed in the $e_{aq}^- + \text{dye}$ reactions (Wojnárovits and Takács, 2008). The lifetime of the intermediate is practically the same in the slightly acidic to slightly alkaline range. However, above pH 10 a short lived intermediate is observed with absorbance maximum at 475 nm. After the decay of this short-lived absorbance the spectrum is similar to the spectrum observed at lower pH (Fig. 8).

At lower pH (3–7) the dye is protonated and the hydrazo form dominates. When the hydrated electron reacts with the molecule we obtain in one step the hydrazo radical; in the radical the negative charge is localized on the oxygen atom and the hydrogen bond is in the $-\text{N}-\text{H}-\text{O}^-$ unit (Scheme 6). Slightly above pH 10 one of the $-\text{OH}$ groups is ionized. In the electron scavenging azo

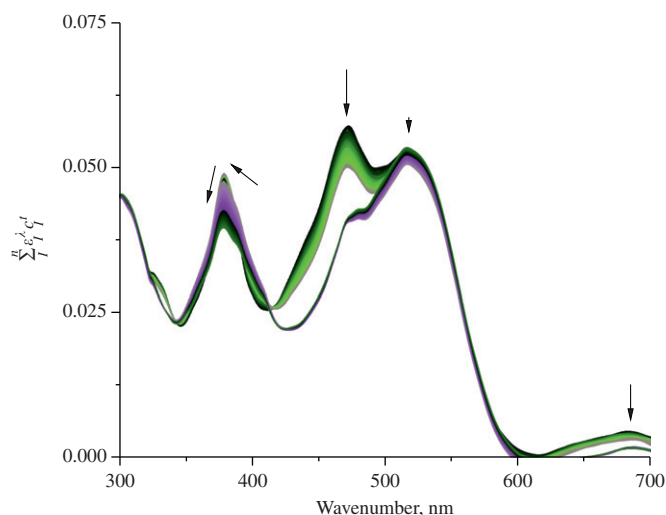


Fig. 8. Absorption spectra of intermediates formed in $e_{aq}^- + \text{SPADNS}$ reaction in the 0–9 ms time range: pH=10.8, $c=0.5 \text{ mmol dm}^{-3}$, $D=10 \text{ Gy imp}^{-1}$. The spectra were generated by using the experimental absorbances and Eq. (1) for correction followed by smoothing.

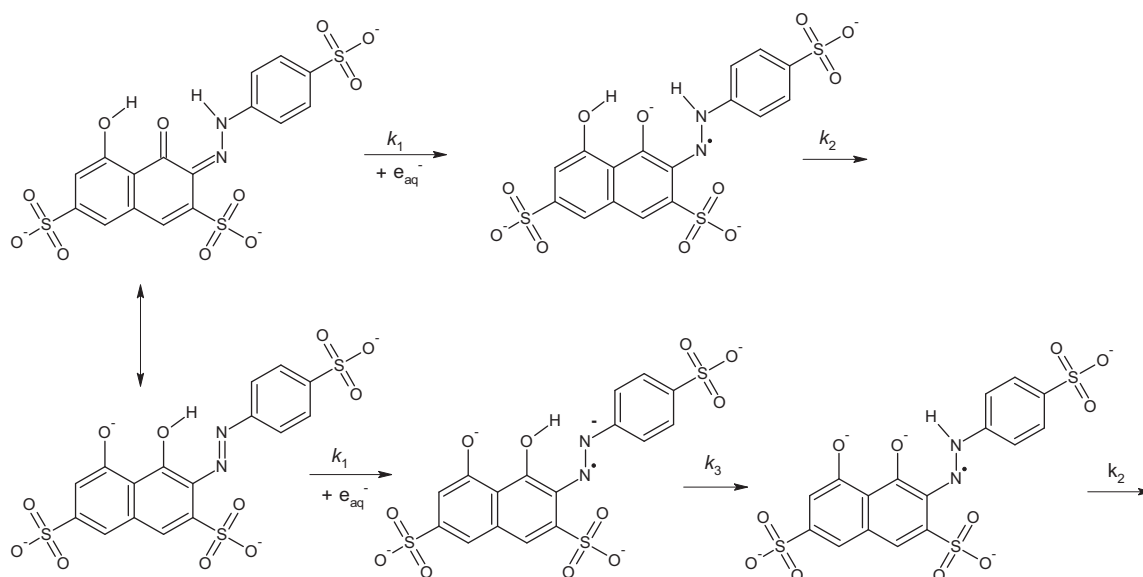
radical anion forms, this radical anion according to the decay of absorbance has a lifetime of $77 \mu\text{s}$, we assume it disappears in transformation to hydrazo radical.

Here also, as with the $\text{HO}^\bullet$ radical reactions, we investigated the decay of intermediates through the formation of stable final products. The absorbance of SPADNS solutions in the visible range was found to decrease strongly upon irradiation without any change in the shape of the spectrum in the visible range. The spectra of the final products as obtained after HPLC separation considerably differ from that of SPADNS: the absorbance maxima of the products are below 400 nm (Fig. 9). On the chromatograms we did not find any compound which has similar absorption spectrum as the starting SPADNS, but its retention time is different from that of the starting compound. The lack of absorbance above 400 nm suggests that the products do not have azo bond. Irradiating $5 \times 10^{-3} \text{ mol dm}^{-3}$ SPADNS solution with 5 kGy dose the absorbance above 400 nm becomes practically zero, the starting molecule completely decomposes. When the dose was increased to 20 kGy the primarily formed stable products also degraded (Fig. 10).

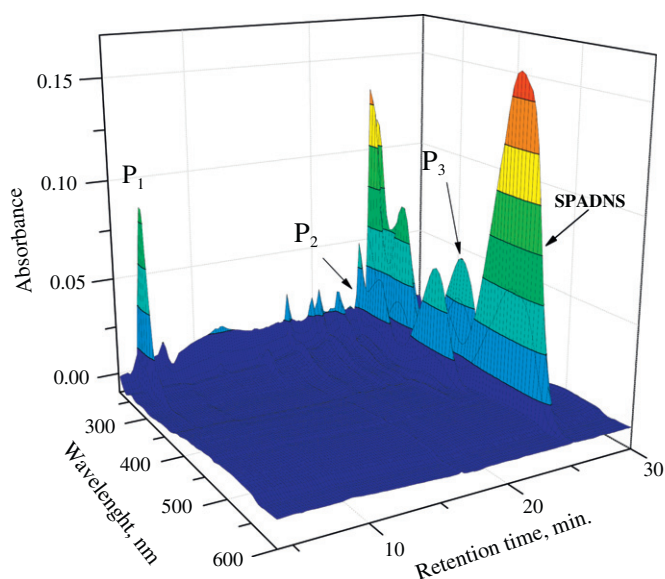
The f -value of the starting molecule disappearance (0.94, Table 4) in the case of SPADNS reveals an irreversible electron scavenging, $k_{-1} \ll k_2$ and k_3 (Scheme 6). The hydrated electron reaction causes in nearly all cases degradation of the molecule and decrease of absorbance in the visible range. We assume that due to the strong decoloration with SPADNS the $(-\text{N}(\text{H})-\text{N}^\bullet-)$ intermediate decomposes with the rupture of $\text{N}-\text{N}$ bond. The absence of molecule III among the radiolysis products of SPADNS supports the idea that the $\text{N}-\text{N}$ bond decomposes and not the $\text{C}-\text{N}$ bond. Formation of III would be a direct proof of $\text{C}-\text{N}$ decomposition at the naphthalene unit.

In the cases of II, III and H-acid molecules, however, the efficiency is much smaller than 1, with II it is smaller than with the other two compounds. In pulse radiolysis investigations the smallest intermediate formation rate and the highest LUMO orbital energy was found for compound II. We assume that the rate coefficient of back reaction (Scheme 5, k_{-1}) is the highest with this compound.

The degradation rate of all compounds studied is higher in $\text{HO}^\bullet$ reactions (Table 3) than in e_{aq}^- reactions (Table 4). The efficiency of degradation (f) in $\text{HO}^\bullet$ reactions for the two simpler molecules is higher than in e_{aq}^- reactions (Table 3), for SPADNS the efficiency is lower in $\text{HO}^\bullet$ reactions than in e_{aq}^- reactions. The higher rate of transformation to intermediates but lower efficiency, in $\text{HO}^\bullet$



Scheme 6. Hydrated electron reaction with SPADNS.

Fig. 9. HPLC 3D chromatogram of irradiated SPADNS solution, $c = 5 \times 10^{-3} \text{ mol dm}^{-3}$, dose 0.8 kGy.

reaction as compared with e_{aq}^- reactions, is due to the higher yield of $\text{HO}^\bullet$ radicals than e_{aq}^- , $5.6 \times 10^{-7} \text{ mol dm}^{-3} \text{ Gy}^{-1}$ (in N_2O saturated solution) and $2.8 \times 10^{-7} \text{ mol dm}^{-3} \text{ Gy}^{-1}$ (N_2 saturated solution *tert*-butanol added), respectively.

4. Conclusions

Sulfanilic Acid Azochromotrop (4,5-dihydroxy-3-(4-sulfophenylazo)naphthalene-2,7-disulfonic acid, SPADNS) as model azo dye and three naphthalene-2,7-disulfonic acid derivatives (without the sulfophenylazo part) were found to decompose efficiently in reaction with $\text{HO}^\bullet$ radicals and hydrated electrons (e_{aq}^-).

- In $\text{HO}^\bullet$ radical induced reactions with the simpler molecules and SPADNS hydroxycyclohexadienyl type radicals form in radical addition to the naphthalene unit. These radicals have characteristic bands in the transient absorption spectra observed in pulse

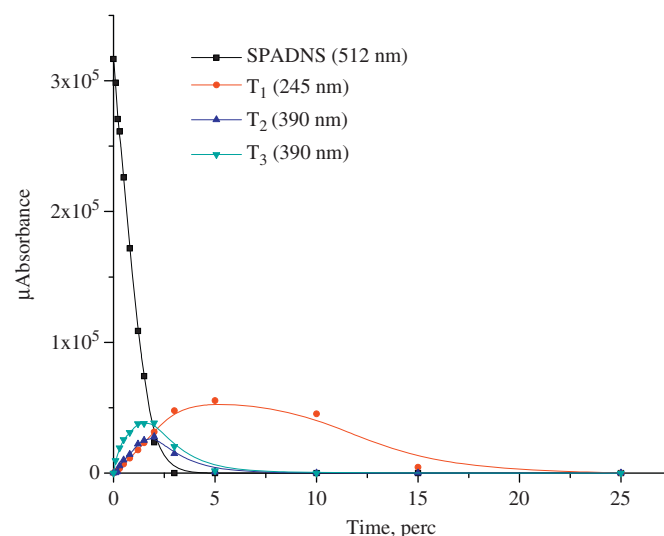
Fig. 10. Product formation in the irradiation of SPADNS under reducing conditions. $c = 5 \times 10^{-3} \text{ mol dm}^{-3}$.

Table 4

Degradation rates (r and r') and efficiencies in *t*-BuOH containing, N_2 saturated solution at pH 5.5.

	r' ($\text{mol dm}^{-3} \text{ Gy}^{-1}$)	r ($\text{mol dm}^{-3} \text{ s}^{-1}$)	f
II	0.64×10^{-7}	0.54×10^{-7}	0.23
III	1.7×10^{-7}	1.4×10^{-7}	0.59
H-acid	1.45×10^{-7}	1.2×10^{-7}	0.51
SPADNS	2.63×10^{-7}	2.19×10^{-7}	0.94

radiolysis. The radical addition partly destroys the extensive conjugation in SPANDS molecule resulting in a decrease in the colour. After the radical addition some of the radicals produced may transform to naphthoxy radical by water elimination.

- In the secondary reactions of hydroxycyclohexadienyl radicals the conjugated electron system giving the colour is partly re-established. However, the absorption spectra of the conjugated products are slightly different (shifted to the red) from that

of the starting molecules. This is due to the hydroxyl substituents formed in the addition reaction. The naphthoxy radicals may dimerize and the dimer molecules have also absorption spectra shifted to the red. The efficiency of degradation in $\text{HO}^\bullet$ reactions is about 0.6–0.7.

- The e_{aq}^- in the reactions with the simpler molecules (without azo bond) adds to the aromatic rings and the anion protonates in fast reaction forming cyclohexadienyl radical. In reaction with SPADNS, e_{aq}^- adds to the azo bond resulting in the destruction of the conjugation through this bond. The decomposition efficiency is close to 1 for SPANDS, while with the simpler molecules it is 0.2–0.6.

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References

- Abbott, L.C., Batchelor, S.N., Oakes, J., Gilbert, B.C., Whitwood, A.C., Smith, J.R., Moore, J.N., 2005. Experimental and computational studies of structure and bonding in parent and reduced forms of the azo dye Orange II. *Journal of Physical Chemistry A* 109 (12), 2894–2905.
- Arslan-Alaton, I., Olmez-Hanci, T., Gursay, B.H., Tureli, G., 2009. $\text{H}_2\text{O}_2/\text{UV-C}$ treatment of the commercially important aryl sulfonates H-, K-, J-acid and Para base: assessment of photodegradation kinetics and products. *Chemosphere* 76 (5), 587–594.
- Bredereck, K., Schumacher, C., 1993. Structure reactivity correlations of azo reactive dyes based on H-acid. I. NMR chemical shift values, pK_a values, dyestuff aggregation and dyeing behavior. *Dyes and Pigments* 21 (1), 23–43.
- Földiák, G., Hargittai, P., Kaszanyiczki, L., Wojnárovits, L., 1988. A computer controlled pulse radiolysis laboratory. *Journal of Radioanalytical and Nuclear Chemistry, Articles* 125 (1), 19–28.
- Földváry, Cs.M., Wojnárovits, L., 2009. Role of reactive intermediates in the radiolytic degradation of Acid Red 1 in aqueous solution. *Radiation Physics and Chemistry* 78 (1), 13–18.
- Frisch, M.J., Trucks, G.W., Schlegel, H.B., Scuseria, G.E., Robb, M.A., Cheeseman, J.R., Montgomery, J.A., Vreven, J.T., Kudin, K.N., Burant, J.C., Millam, J.M., Iyengar, S.S., Tomasi, J., Barone, V., Mennucci, B., Cossi, M., Scalmani, G., Rega, N., Petersson, G.A., Nakatsuji, H., Hada, M., Ehara, M., Toyota, K., Fukuda, R., Hasegawa, J., Ishida, M., Nakajima, T., Honda, Y., Kitao, O., Nakai, H., Klene, M., Li, X., Knox, J.E., Hratchian, H.P., Cross, J.B., Bakken, V., Adamo, C., Jaramillo, J., Gomperts, R., Stratmann, R.E., Yazyev, O., Austin, A.J., Cammi, R., Pomelli, C., Ochterski, J.W., Ayala, P.Y., Morokuma, K., Voth, G.A., Salvador, P., Dannenberg, J.J., Zakrzewski, V.G., Dapprich, S., Daniels, A.D., Strain, M.C., Farkas, O., Malick, D.K., Rabuck, A.D., Raghavachari, K., Foresman, J.B., Ortiz, J.V., Cui, Q., Baboul, A.G., Clifford, S., Cioslowski, J., Stefanov, B.B., Liu, G., Liashenko, A., Piskorz, P., Komaromi, I., Martin, R.L., Fox, D.J., Keith, T., Al-Laham, M.A., Peng, C.Y., Nanayakkara, A., Challacombe, M., Gill, P.M.W., Johnson, B., Chen, W., Wong, M.W., Gonzalez, C., Pople, J.A., 2002. Gaussian 98. Gaussian, Inc., Pittsburgh PA.
- Han, B., Ko, J., Kim, J., Kim, Y., Chung, W., Makarov, I.E., Ponomarev, A.V., Pikaev, A.K., 2002. Combined electron-beam and biological treatment of dyeing complex wastewater. Pilot plant experiments. *Radiation Physics and Chemistry* 64 (1), 53–59.
- Hayon, E., Treinin, A., Wilf, J., 1972. Electronic spectra, photochemistry, and autoxidation mechanism of the sulfite-bisulfite-pyrosulfite systems. The SO_2^- , SO_3^- , SO_4^- , and SO_5^- radicals. *Journal of the American Chemical Society* 94 (1), 47–57.
- Inskeep, W.H., Jones, D.L., Silfvast, W.T., Eyring, E.M., 1968. Intramolecular hydrogen bonding of azo dyes in aqueous solution. *Proceedings of the National Academy of Sciences of the United States of America* 59 (4), 1027–1029.
- Johnston, L.J., Mathivanan, N., Negri, F., Siebrand, W., Zerbetto, F., 1993. Assignment and vibrational analysis of the 600 nm absorption band in the phenoxyl radical and some derivatives. *Canadian Journal of Chemistry* 71 (10), 1655–1662.
- Neta, P., Madhavan, V., Zemel, H., Fessenden, R.W., 1977. Rate constants and mechanism of reaction of SO_4^{2-} with aromatic compounds. *Journal of the American Chemical Society* 99 (1), 163–164.
- Özen, A.S., Aviyente, V., Klein, R.A., 2003. Modeling the oxidative degradation of azo dyes: a density functional theory study. *Journal of Physical Chemistry A* 107 (24), 4898–4907.
- Pálfi, T., Takács, E., Wojnárovits, L., 2007. Degradation of H-acid and its derivative in aqueous solution by ionising radiation. *Water Research* 41 (12), 2533–2540.
- Paul, J., Naik, D.B., Sabharwal, S., 2010a. High energy induced decoloration and mineralization of Reactive Red 120 dye in aqueous solution: a steady state and pulse radiolysis study. *Radiation Physics and Chemistry* 79 (7), 770–776.
- Paul, J., Naik, D.B., Sarkar, S.K., Sabharwal, S., 2010b. One electron oxidation and reduction of Reactive Red-120 dye in aqueous solution: a steady state and pulse radiolysis study. *Radiation Physics and Chemistry* 79 (12), 1225–1233.
- Peintler, G., Kormányos, B., Gyurcsik, B., 2007. pHCall. Szeged, <<http://www.staff.u-szeged.hu/~peintler/huprogs.htm#phcall>>.
- Press, W.H., Flannery, B.P., Teukolsky, S.A., Vetterling, W.T., 1992. Runge–Kutta method and adaptive step size control for Runge–Kutta. In *Numerical recipes in FORTRAN: The Art of Scientific Computing*, second ed. Cambridge University Press, Cambridge, England, pp. 704–716.
- Roder, M., Wojnárovits, L., Földiák, G., Emmi, S.S., Beggiato, G., D'Angelantonio, M., 1999. Addition and elimination kinetics in OH radical induced oxidation of phenol and cresols in acidic and alkaline solutions. *Radiation Physics and Chemistry* 54 (5), 475–479.
- Steenken, S., Neta, P., 2003. Transient phenoxyl radicals: formation and properties in aqueous solution. In: Rappoport, Z. (Ed.), *Chemistry of Phenols*. Wiley, Chichester, pp. 1107–1152.
- Swallow, A.J., 1982. Application of pulse radiolysis to study of aqueous organic systems, in: The study of fast processes and transient species by electron pulse radiolysis. In: Baxendale, J.H., Busi, F. (Eds.), *Proceedings of the NATO Advanced Study Institute, Capri, 1981*, Reidel, Dordrecht, pp. 289–315.
- Von Sonntag, C., 2006. Free radical induced DNA damage and its repair: a chemical perspective. Springer Verlag, Heidelberg.
- Wojnárovits, L., Pálfi, T., Takács, E., 2005. Irradiation treatment of textile dyes: Apollon Red. *Research on Chemical Intermediates* 31 (7–8), 679–690.
- Wojnárovits, L., Pálfi, T., Takács, E., 2007. Kinetics and mechanism of azo dye destruction in advanced oxidation processes. *Radiation Physics and Chemistry* 76 (8–9), 1497–1501.
- Wojnárovits, L., Takács, E., 2008. Radiation induced degradation of azo dyes: an overview. *Radiation Physics and Chemistry* 77 (3), 225–274.
- Zhang, S.-J., Yu, H.-Q., Li, Q.-R., 2005. Radiolytic degradation of Acid Orange 7: a mechanistic study. *Chemosphere* 61 (7), 1003–1011.
- Zhurko, G., 2008. <<http://www.downloadsoftfree.com/windows/Home-Education/Miscellaneous/Chemcraft-Linux-academic-license-1-4-57-72805-1-0-0.html>>.